

We consider

$$f(x_1, x_2) = 2x_1^2 + 9x_2^2.$$

The second derivative matrix is

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} 4 & 0 \\ 0 & 18 \end{pmatrix}.$$

1. The matrix is diagonal. Therefore, its eigenvalues are obtained directly as the diagonal entries. Consequently, the condition number is

$$\kappa_2(A) = \frac{18}{4} = \frac{9}{2},$$

for any $x \in \mathbb{R}^2$.

2. We now apply the change of variables

$$\begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3\sqrt{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{\sqrt{2}}{6} \end{pmatrix} \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix}$$

We obtain

$$\begin{aligned} \tilde{f}(x'_1, x'_2) &= f\left(\frac{1}{2}x'_1, \frac{\sqrt{2}}{6}x'_2\right) \\ &= 2\left(\frac{1}{2}x'_1\right)^2 + 9\left(\frac{\sqrt{2}}{6}x'_2\right)^2 \\ &= \frac{1}{2}x'^2_1 + \frac{1}{2}x'^2_2. \end{aligned}$$

The Hessian of $\tilde{f}$ is the identity matrix:

$$\nabla^2 \tilde{f}(x') = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore, the two eigenvalues are equal to 1, and the condition number is 1, for any $x' \in \mathbb{R}^2$.